

Conclusions — Comparison with Published Results

****Primary finding fully replicated:**** SCF1 (V3 B9J08_03708) is the most significantly downregulated gene in both comparisons, at $\log_2FC = -6.82$ (tnSWI1 vs AR0382) and -7.35 (AR0387 vs AR0382), both with $padj \approx 0$. This matches Fig. 1D and Fig. S5A of the paper quantitatively.

The paper states: 'The strongest, most significantly dysregulated gene in tnSWI1 was [SCF1]' and 'in the poorly adhesive AR0387, SCF1 was the most down-regulated gene compared to AR0382.' Both statements are confirmed by our reanalysis using the newer V3 genome assembly.

****Gene ID discrepancy resolved:**** V3 locus tags use 5-digit numbering (B9J08_03708) vs V2's 6-digit format (B9J08_001458). Without protein-level matching, SCF1 cannot be found by ID alone. Diamond blastp RBH produced 4,567 high-confidence V3↔V2 pairs ($\geq 50\%$ amino acid identity).

****Quantitative agreement:**** Our fold-changes (-6.82 and -7.35) are consistent with the ~ 29 -fold difference reported in the paper between AR0382 and AR0387 SCF1 expression ($\log_2(29) \approx 4.9$).

Limitations & Notes

- Only 2 biological replicates per condition → reduced statistical power for marginal genes
- Higher DE gene counts vs paper (1,234 / 1,582 vs paper's estimates) likely reflect V3 gene model differences and possible batch effects not modelled
- Strandedness inferred from kit documentation — not verified computationally (e.g. via RSeQC)
- 857 V3 genes had no V2 Diamond hit; these cannot be named using V2 gene nomenclature
- MultiQC alignment statistics not yet reviewed (rnaseq-pe workflow still completing)

Analysis Cost

Component	Description	Cost (USD)
Galaxy compute	STAR alignment × 6 samples, featureCounts, DESeq2 × 2, Diamond × 2	\$0.00
LLM API calls	claude-sonnet-4-6 via Anthropic API (planning, execution, interpretation)	\$24.87
Data transfer	SRA download, NCBI FTP (genome/proteins), UCSC GTF	\$0.00
Total	Complete RNA-seq reproduction from raw FASTQs to publication figures	\$24.87

* usegalaxy.org is a free public resource. Galaxy compute costs are covered by NSF/NIH grants to the Galaxy Project. Commercial cloud equivalent estimated at \$3–8 for this workload.
Report generated: 2026-06-02 | Reference: Santana & O'Meara, Science 381:1461 (2023) | BioProject: PRJNA904261